

Kennesaw State University

Department of Computer Science

CS 8267

RESEARCH CRITICS: Advanced Machine Learning

This assignment requires you to critically analyze a recent and influential research paper in the field of machine learning. You will evaluate its strengths, weaknesses, contributions, and compare it with other relevant works including newer ones.

Goals

- Study one of the advanced research topics in machine learning with critical thinking
- Learn how well-written research papers are structured and presented, so you can effectively craft and communicate your own research ideas.

You may complete this assignment individually or in a group of up to two members. The expectations and grading criteria are the same for both formats. A maximum of 16 research critics will be accepted. If the class has more than 16 students, group formations will be allowed to ensure everyone can participate.

Group Project Policy

1. You are responsible for choosing group members. While choosing your group, consider the following:
 - a. Schedules of your group members
 - b. Interest in the paper or topic
2. All group members should participate at all levels of the research critics.
3. If a student member does not contribute, please let me know ASAP.

Tasks

- Find a research paper in the area of machine learning that has been published in the last 5 years.
 - Minimal requirement:
 - The paper must have at least **200** citations.
 - Domain-specific (e.g., medical domain) papers need approval.
 - The paper should be published in 2017 or later.

- Papers from following conferences and journals are acceptable without citation requirement.
 - ICML(International conference on Machine Learning),
NeuroIPS (Conference on Neural Information Processing Systems),
ICLR (International Conference on Learning Representations), CVPR
(IEEE/CVF Conference on Computer Vision and Pattern Recognition),
IEEE Transactions on Artificial Intelligence, IEEE Transactions on
Pattern Analysis and Machine Intelligence, Neural Networks
- Review and survey papers are not acceptable.
- Read and understand the paper carefully.
- Explain
 - the motivation of the paper
 - 3 strengths of the paper
 - 1 weakness of the paper
 - Hint: A new paper citing the paper would normally explain its weakness.
 - the major contribution of the paper (how does the paper advance the field?)
- Explain datasets used in the paper: size, structure, availability, etc.
- Explain quantified comparisons made with other techniques.
- Evaluate the thoroughness of experiments.
- Discuss whether the authors cited proper relevant work in the area.
- Discuss what authors could have done in addition to described in the paper to strengthen their paper.
- Provide a rating for the following in the range of [1-5] (5 is highest).
 - Title
 - Abstract
 - Introduction
 - Related Work
 - Method Explanation
 - Experiments
 - Conclusion
 - Overall rating

Deliverables

- 2-page single-spaced report using 11-point Times New Roman font. Use proper section headers and title.
- 7-minute presentation that explains the paper and summary of your report including 2-minute Q&A at the end of your presentations.

These intermediate dates are very important.

LATE SUBMISSION

You will lose 1% per week (from report and presentation) if you miss these intermediate deadlines.

Important Dates

Group notification:

September 2nd, 2026

Paper selection:

September 21st, 2026

Report:

October 26th, 2026

Presentations:

October 26th and 28th, 2026